Afshin Malek

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TECHNICAL SUMMARY

Systems and software engineering graduate student, 4.0 GPA across 45 units of advanced STEM coursework, proficient in C++ and Java systems programming, data structures, concurrency, file I/O, and MERN full-stack development.

TECHNICAL SKILLS

Languages: C++, C, Java, Python, JavaScript (ES6+), x86 / MIPS Assembly, SQL, Bash, HTML5, CSS3

Systems & Architecture: Object-Oriented Design, Data Structures & Algorithms, Computer Architecture, Memory Management, Linux/POSIX CLI, File I/O Streams, Concurrency, Networking

Web & Frameworks: React, Redux Toolkit, React Router, Node.js, Express.js, MongoDB, Mongoose, RESTful APIs, JSON, Axios, Nodemailer, Passport

Developer Tools: Git, GitHub, Make/CMake, VS Code, GDB

Mathematics: Discrete Mathematics, Linear Algebra, Multivariable Calculus

EDUCATION

California State University, East Bay Aug 2026 - Present

Graduate Computer Science Bridge (Conditional M.S.)

Upper-division Data Structures and Algorithms, Computer Organization and Architecture; Spring 2027: Operating Systems, Analysis of Algorithms; target Georgia Tech OMSCS Computing Systems Fall 2027.

Peralta Community College District Aug 2024 - May 2026

Post-Baccalaureate CS & Mathematics

GPA 4.00/4.00, 45 units, Dean's List, Certified Peer Tutor.

San Francisco State University Jan 2016

B.S. Business Administration (Finance)

GPA 3.34.

WORK EXPERIENCE

Computer Science & STEM Peer Tutor Jul 2025 - May 2026

Berkeley City College, Berkeley, CA

- Structured tutoring in C++ and Java OOP, recursion, pointers, data structures, Calculus, Linear Algebra.
- One-on-one code reviews and live debugging: compiler diagnostics, memory management, clean code.

Full-Stack Web Developer Jan 2023 - Jun 2025

Self-Employed (Freelance), San Francisco Bay Area, CA

- MERN full-stack web apps (MongoDB, Express.js, React, Node.js): responsive UI, routing, state management.
- Secure RESTful APIs with JSON and authentication; MongoDB schema design; Git/GitHub workflows.

TECHNICAL PROJECTS

1. Battleships Java, Swing, Networking

github.com/afshinmalek/battleships

Socket-based client/server Battleships with 25x25 player and tracking grids, deterministic ship placement, and synchronized turn state. Hit/miss/sunk/win message protocol for shot resolution and grid updates between client and server.

2. BookMe - MERN Stack Full-Stack Applications React, Redux Toolkit, Express.js, Node.js, MongoDB, REST APIs

github.com/afshinmalek/bookedCRA-client | github.com/afshinmalek/bookedCRA-server

Component-driven React single-page application with Redux Toolkit state management, React Calendar booking, and client-side routing; modular Express/Node.js services exposing CRUD RESTful endpoints. Google OAuth via Passport; MongoDB with Mongoose modeling and a unique date + time index that blocks double-bookings; Nodemailer confirmation emails.

3. C++ Spaced Repetition Study Engine C++, OOP, File I/O, CLI

github.com/afshinmalek/spaced-repetition-quiz-engine

Streak-based mastery scheduling: low-streak questions repeat, a streak of 2+ means mastered; encapsulated class hierarchies, file-stream persistence, and strict input validation.

4. Red-Black BST Java, Data Structures

github.com/afshinmalek/red-black-bst

Generic self-balancing red-black binary search tree with logarithmic insertion, deletion, and in-order traversal. Tested by inserting 1-60 and deleting 1-20.

5. Dijkstra Shortest Path Java, Graphs, Priority Queues

github.com/afshinmalek/dijkstra-shortest-path

Dijkstra's shortest-path algorithm over an indexed min-priority queue on a 50-node weighted graph. Assigns each customer to the nearest of field stations 7, 14, 25, 28, and 40 and prints the route.

6. Lazy Prim's MST - Minimum Cable Network Java, Graphs

github.com/afshinmalek/lazy-prim-mst

Lazy Prim's minimum spanning tree for cable-network optimization, with per-station MST costs to choose the cheapest field station.

7. Fork/Join Distance Calculator Java, Concurrency

github.com/afshinmalek/fork-join-distance-calculator

Fork/Join parallelism benchmarked against single-threaded execution on 10 million point pairs. Performance compared at several split thresholds.

8. File Processing Utilities C++, File I/O, Stream Parsing

github.com/afshinmalek/file-processing

20 single-pass streaming utilities built on ifstream and stringstream, parsing CSV, multi-delimiter, key=value configuration, fixed-width, and quoted-field records without whole-file loading. Header-aware column aggregation, predicate-based filtering, and extremum searches, with runtime_error handling for missing files and columns.

9. Supplier-Consumer Threads Java, Concurrency

github.com/afshinmalek/supplier-consumer-threads

Ten suppliers and ten consumers coordinated through a shared MeetingRoom using synchronized wait/notify. Timestamped logging of supplier and consumer activity.

10. Game of Life Java, Swing

github.com/afshinmalek/game-of-life

Swing GUI for Conway's Game of Life with a threaded generation loop. User-chosen grid size for the simulation.